

1. This element of music is measured in Hertz. Some instruments, including drums, cymbals, and triangles, do not have a definite one. Name this element that determines how high or low a note is.

PITCH

2. In this novel, Lily hopes to return jars full of stories to save her Korean grandmother's life. She repeatedly sees the title animal, who tells her stories after being captured. Name this recent Newbery Medal winner by Tae Keller.

WHEN YOU TRAP A TIGER

3. This is one of the properties of real numbers. What property of addition states that if there are more than two addends, it doesn't matter which two addends are added first?

ASSOCIATIVE (PROPERTY OF ADDITION)

4. Traditionally, this property of a substance consists of length, width, and height. Its SI unit is the cubic meter, but English measures like the gallon are still prevalent in the US. Give the word for this measurement of how much space matter takes up.

VOLUME

5. The four Kentucky counties which border this body of water include Ballard, Carlisle, Fulton, and Hickman. What major river forms the boundary between western Kentucky and Missouri?

MISSISSIPPI (RIVER)

6. Mrs. Jenkins is filling treat bags for the academic team. She needs 4 Tootsie Pops for each bag. If there are 15 members on the team, how many pops will she need?

60 (SIXTY) (TOOTSIE ROLL POPS)

7. In this picture book, a young girl helps a wild pup find his way back to his pack. In return, the pack helps the girl's parents locate her. Name this book that earned Matthew Cordell a Caldecott.

WOLF IN THE SNOW

8. This Republican lawmaker announced his campaign for president of the United States in 2015, a decision made during his first term as the junior US senator for Kentucky. Identify this politician who was re-elected as one of Kentucky's two US senators in 2022.

(RAND) PAUL

9. This relative of country and folk music features the fiddle, mandolin, and banjo. Its pioneers included Lester Flatt, Earl Scruggs, and Bill Monroe. What is this genre that originated in Appalachia?

BLUEGRASS

10. This force converts some kinetic energy into heat energy. Lubricant reduces it to improve a machine's efficiency. Give the name for this resistance to motion when one object rubs against another.

FRICTION

11. This SI unit honors the Scottish perfecter of the steam engine. It equals one joule per second and is calculated by multiplying a circuit's amps by its volts. What is this measurement of power?

WATT

12. This instrument comes in lever and pedal varieties. It has a wooden triangular frame and, depending on the model, between 12 and 47 strings. Name this large instrument associated with angels.

HARP

13. The lines used for this type of geographic coordinate are called meridians. Give the term for the system of measuring a point's position either east or west of the prime meridian.

LONGITUDE

14. Edward Lear popularized this short kind of poem in the 1800s. It is five lines long and has an AABBA rhyme scheme. What is this funny poem that takes its name from an Irish city?

LIMERICK

15. It takes Jeffrey $\frac{3}{5}$ of an hour to walk to school. How many minutes does it take him to walk to school?

36 (THIRTY-SIX MINUTES)

16. This instrument is pitched an octave below the trumpet in C or B-flat. It has the same range as the baritone horn. What is this low brass instrument that serves as a tenor for the tuba?

EUPHONIUM [yoo-FO-nee-um]

17. In this novel, three youngsters tesser to Camazotz, where an enormous brain called IT controls all individuals. Calvin O'Keefe helps Meg and Charles Wallace Murry rescue their scientist father and bring him back to Earth. Identify this Newbery-winning Madeleine L'Engle novel.

(A) *WRINKLE IN TIME*

18. This sedimentary rock composes more than half of Kentucky's surface rocks. Made from calcite, it is used in glass and cement production. Name this rock that consists of animals' organic remains, such as coral and shells.

LIMESTONE

19. Along with beans and squash, this crop forms the "Three Sisters," which is vital to many indigenous peoples. Cultivation of what cereal grain was successfully introduced to the colonists of Plymouth by Squanto?

CORN (ACCEPT: MAIZE)

20. A museum contains 12 impressionist paintings and 15 landscapes. What is the ratio of impressionist paintings to the total number of paintings?

4:9 (FOUR TO NINE) (ACCEPT: FOUR OVER NINE OR FOUR-NINTHS)

21. This genre is sometimes confused with the memoir. *Boy* by Roald Dahl [ROO-all DOLL] and *I Am: LeBron James* fall into this category. Identify this type of book in which a person writes their life story.

AUTOBIOGRAPHY

22. While this early English settler is often credited for introducing tobacco cultivation to English colonies, he is perhaps best remembered for his marriage to the daughter of the Native American chief Powhattan. Name this colonist who, in 1614, married Pocahontas.

(JOHN) ROLFE

23. Among this man's more than 600 songs are "Gretchen at the Spinning Wheel" and "The Erl King." Like Beethoven, he bridged the Classical and Romantic Eras. Whose Eighth Symphony was nicknamed "Unfinished"?

(FRANZ) SCHUBERT

24. This two-word measurement indicates atmospheric humidity. It is the temperature at which water vapor saturates the atmosphere. Give the term for the temperature at which water begins to condense from the air.

DEW POINT

25. A road measures 9400 meters long. The second road is 600 meters long. What is the combined length of the roads in kilometers?

10 (TEN KILOMETERS)

26. A regular polygon with four sides is drawn on the board. Each angle equals 90 degrees, and each side is 12 inches long. What polygon was drawn on the board?

SQUARE

27. As a baby, this tall tale character teathed on a bowie knife and played with wild animals. After he fell out of his parents' wagon, coyotes raised him near a Texas river. Name this cowboy who, for a very brief time, was the husband of Slue-Foot Sue.

PECOS BILL

28. This American university was founded in 1636 in Cambridge, Massachusetts. Name the United States' oldest university, a school known as one of the most prestigious schools in the world.

HARVARD (UNIVERSITY)

29. This dance comes in New York, Miami, and LA styles in continental North America. In South America, Colombia adopted this dance, which originated in Cuba. What is this dance that shares its name with a tomato, onion, and pepper hot sauce?

SALSA

30. Latin name, please. This constellation helped many enslaved Americans escape north to freedom. Seven of its brightest stars form the Big Dipper. Identify this constellation, whose name translates to "great bear."

URSA MAJOR

End of First Half – Begin Three-Minute Inquiry Period

Round 2 - Second Half

Elementary – 2023 District



31. Certain eukaryotic [yoo-kare-ee-AH-tik] cells feed or move through this extension of cytoplasm. It is a temporary arm-like projection that protozoans, especially amoebas, use. Identify this so-called "false foot."

PSEUDOPOD [SOO-duh-pod] (Accept PSEUDOPODIA, PSEUDOPODIUM)

32. With her sister Marjorie, this dancer founded the Chicago City Ballet in 1980. She performed with the Ballet Russe [ROOS] de Monte Carlo and for her husband George Balanchine [BAL-un-sheen] with the New York City Ballet. Give the name of this first prima ballerina of Native American descent.

(MARIA) TALLCHIEF

33. Thanks to its alliteration and consonance, this nursery rhyme also serves as a tongue twister. In it, the title character harvests 1/4 of a bushel's worth of a particular vegetable. Give the name of this rhyme.

"PETER PIPER (PICKED A PECK OF PICKLED PEPPERS)"

34. Forces led by William Henry Harrison defeated the brother of this Shawnee leader at the Battle of Tippecanoe. Harrison's forces later killed him at the Battle of the Thames [temz]. Give the name of this Shawnee chief who joined forces with the British to capture Detroit during the War of 1812.

TECUMSEH

35. Give your answer in terms of pi. A pizza pan has a radius of four units. What is the circumference of the pizza pan?

8 PI (EIGHT PI UNITS)

36. Before this theatrical job, actors or playwrights were responsible for mounting a production. Today, a person with this title guides the cast and crew through rehearsals but usually lets a stage manager lead the performances. Identify this person who integrates staging, lighting, costume, props, sound, and acting into a finished play or musical.

DIRECTOR

37. Correct the following sentence, if needed: Me and Taylor went to the concert.

CHANGE "ME AND TAYLOR" TO "TAYLOR AND I" (Accept equivalents)

38. You roll two fair dice. How many possible ordered pair combinations could you roll?

36 (COMBINATIONS)

39. This process begins when chlorophyll absorbs solar energy. This light energy transforms carbon dioxide and water into oxygen and sugars. What is this process that green plants use to make their food?

PHOTOSYNTHESIS

40. In one of this US president's best-remembered speeches, he said, "Mr. Gorbachev, tear down this wall." Who served as the United States' 40th president from 1981 to 1989?

(PRESIDENT RONALD) REAGAN

41. The first part of this compound word is the wooden core of an ear of corn. The second half follows "World Wide" in the Internet's original name. Identify this word for the silken home a spider spins.

COBWEB

42. Recent holders of this Kentucky government office have included Crit Luallen, Jenean Hampton, and Jacqueline Coleman. The holder of what position is first in the line of succession should Kentucky's governor die or be removed from office?

LIEUTENANT GOVERNOR

43. Algebraic expressions and numerical expressions differ because algebraic expressions contain these. What is included in an algebraic expression that is not included in a numerical expression?

VARIABLE (ACCEPT: LETTERS LIKE X AND Y OR SIMILAR RESPONSE)

44. In butterflies, this stage of complete metamorphosis is called the chrysalis [KRIS-uh-lis]. Many moths and caterpillars spin a cocoon to protect it. Name this stage between larva and adult.

PUPA(L) (STAGE)

45. The two-dimensional kind of this artistic element is visual, while the three-dimensional type is tactile. Adjectives that may describe it include "bumpy," "smooth," "rough," and "fuzzy." Give the word for how an artwork feels or looks like it would feel.

TEXTURE

46. M represents money in an algebraic equation. Haven saves M dollars each week. How much money will she save in 15 weeks?

15M (FIFTEEN M) (ACCEPT: 15 TIMES M)

47. This architectural feature is shared by Rome's Pantheon, India's Taj Mahal, and the US Capitol Building. Give the name of this roof or ceiling shaped like an overturned bowl.

DOME(S)

48. This organ comes in dorsal, anal, pectoral, ventral, and tail varieties. Made of bony or soft spines called rays, they help certain aquatic species balance, steer, and protect themselves. Name these appendages that sometimes resemble flippers.

FIN(S)

49. While there is no national form of this type of tax, all but five US states require that this tax be paid when purchasing most goods or services. What is this type of tax that in Kentucky is set at 6%?

SALES (TAX)

50. Which words need to be capitalized in the following sentence? Nick asked, "Will you please come to my birthday party?"

NICK (AND) WILL (either order)

51. This question requires multiple answers. What are the two prime numbers between four and ten?

5, 7 (FIVE, SEVEN)

52. This man decorated a Roman church's altar wall with his fresco, *The Last Judgment*. He also adorned that space with *The Creation of Adam* and other biblical scenes. Identify this Renaissance painter of the Sistine Chapel ceiling.

MICHELANGELO (BUONARROTI)

53. Within a free market, the price of a good or service is determined by this measurement's relationship with an item's demand. Give the term which explains the current availability of a given product or service.

SUPPLY

54. These large divisions are part of the kidneys, liver, lungs, and brain. Give this word that applies to the fleshy section at the bottom of the ear.

LOBE(S)

55. This part of speech might be transitive or intransitive. Linking ones are usually forms of "be" while helping ones express mood and tense. Which part of speech also includes action words?

VERB

56. Only one of these clauses is part of a simple sentence. Give the name for a clause that contains a subject and a verb and expresses a complete thought.

INDEPENDENT (CLAUSE) (Accept MAIN (CLAUSE))

57. This step in the scientific method immediately precedes asking a question. State the scientific method's first step.

(MAKE AN) OBSERVATION (Accept equivalents)

58. It takes 650 milligrams of silver to make one charm. Hannah has 1.4 grams of silver. How many charms can she make?

2 (TWO) (CHARMS)

59. Though this number is sometimes considered unlucky, its importance to the founding of the United States can be seen in the number of stripes on the American flag. Give this number, which equals the number of colonies that declared their collective independence from British rule.

13

60. At his death in 1891, this artist's painting, *The Circus*, was unfinished. His first painting set at night was 1888's *Circus Sideshow*. Identify this man who co-founded Pointillism with Paul Signac [seen-YAHK].

(GEORGES) SEURAT [ZHAWRZH suh-RAH]

End of Second Half – Begin Three-Minute Inquiry Period

Extra Questions

61. Traditionally, men weave this cloth, and women sew the strips together. Once reserved for royalty and made of silk, it is now made of cotton and available to everyone. What is this brightly colored and patterned cloth native to Ghana?

KENTE [KEN-tay] (CLOTH)

62. Spelling required. Most herbs will freeze successfully. Spell the word from the preceding sentence, an antonym of "thaw."

F-R-E-E-Z-E

63. There are two green balls, six red balls, one orange ball, and two yellow balls in a bag. If you choose a ball from the bag without looking, what is the probability that the ball will not be black?

1 (ONE) (ACCEPT: 100 PERCENT)

64. This man was a naturalist on the *HMS Beagle's* expedition along South America's coasts. He scandalized Christians by suggesting that species gradually changed over hundreds of years. Who proposed evolution by natural selection in his book *On the Origin of Species*?

(CHARLES) DARWIN

65. This member of the British royal family served as heir apparent for more than seven decades before he succeeded his mother, Queen Elizabeth the Second, to the throne in 2022. Give the name of this newly crowned British monarch.

(KING) CHARLES (THE THIRD)